Emilija Kastratović

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Personal Profile

Master's student in Computer Science with 4+ years of practical experience in Cybersecurity and Software Engineering who brings a creative and analytical mindset to technical problem-solving.

Professional Experience

Airbus Defence and Space — March 2026 – present
Working Student — AI Tool Development — *Software Engineering*

Hitachi Rail GTS Deutschland — February 2025 – February 2026
Working Student — Penetration Testing — *Cybersecurity · Kali Linux · Penetration Testing*

- Performed penetration testing on railway systems and products.
- Created and maintained custom Kali Linux ISO with automated builds using Jenkins.
- Developed a custom Burp Suite plugin to simulate railway-specific traffic protocols for security testing purposes.

DENSO Automotive Deutschland — October 2023 – November 2024
Working Student — Fundamental Technology R&D — *Apache Kafka · Cybersecurity · Python*

CONNECT Horizon Europe research program focusing on Trust Management in VANETs.

- Implemented, tested, and maintained a Python-based Cooperative Adaptive Cruise Control (CACC) module within SUMO traffic simulations, using Apache Kafka for real-time vehicle communication on Ubuntu Linux.
- Integrated communication between the Trust Assessment Framework (TAF) and partner systems to dynamically migrate CACC instances between ECUs based on Trust Levels.

Mercedes-Benz Tech Innovation — October 2021 – September 2023
Working Student — CarIT Security — *Cybersecurity · Project Management · Python*

- Developed a Python-based tool for automated cybersecurity analysis of vehicular network architectures.
- Performed vulnerability analysis on vehicle systems to identify security weaknesses.
- Assisted with project management tasks, including documentation and review processes.

Bachelor's Thesis

- "Comparing different vehicle architectures based on attack path analysis".

Education

Computer Science, M.Sc. October 2023 – June 2026

Ulm University

- Master's Thesis:
  "Assessing the Effectiveness of Large Language Model Support for Generating GDPR ROPA Documentation"

Software Engineering, B.Sc. October 2019 – July 2023

Ulm University

- Bachelor's Thesis (in cooperation with Mercedes-Benz Tech Innovation):
  "Comparing different vehicle architectures based on attack path analysis": Created criteria and developed a Python tool for assessing vehicular network security by comparing attack paths, and calibrated these criteria through expert surveys and mathematical modeling, enabling automated security evaluations.

Projects

ROPAgen (ropagen.eu) since April 2025

Full Stack Developer *Next.js · TypeScript · Docker · Git · CI/CD · Supabase · OpenRouter · Figma*

- Developed a self-hosted, LLM-driven platform at Ulm University (DBIS) for GDPR compliance, featuring a dual AI focus: automatic generation of ROPA documents and an integrated LLM support via AI suggestions and chatbots, using OpenRouter. Established CI/CD pipelines for automated deployment.

FIND.ME (festival-find.me) May 2025

Full Stack Developer *Next.js · TypeScript · Docker · Git · Supabase*

- Built a digital festival companion web application which allows users to live track their own and their friends' stage location, view all festival acts, mark favorite acts, and manage profiles.

BierTurnier (tournament.beer) November 2023 – December 2024

Full Stack Developer *Next.js · TypeScript · Docker · Git · Supabase · Adobe Photoshop · Adobe Illustrator*

- Created a web application to create and track party games with friends. Designed and implemented the UX/UI, brand identity and visual design, logos, and graphic assets.

Skills

Programming: Python · TypeScript · Java
Frameworks & Libraries: Next.js · React.js · Three.js · Framer Motion · OpenRouter · Apache Kafka · Supabase
Tools & Platforms: Docker · Docker Compose · Git · GitHub · Jenkins · Kali Linux · Burp Suite · Figma · CI/CD · Agile Development · Penetration Testing · Security Engineering · Software Engineering
Languages: German (native) · Serbian (native) · English (C1) · Spanish (B1) · French (B1)